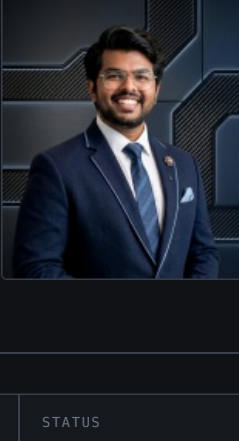


Niraj Naidu

MECHANICAL PRODUCT DESIGNER · HYDRAULIC SYSTEMS · **SOLIDWORKS** / **CATIA** / **ANSYS**

A detail-oriented mechanical product designer with hands-on experience in 3D modeling, hydraulic system design, reverse engineering, and precision machining — delivering solutions from gantry cranes and mold presses to tube-bending machines and Formula Student race cars.

↓ Download CV (PDF) ↓ Website Overview (PDF)



01 / ABOUT

Engineering with precision

Currently pursuing a Master of Science in Digital Technologies at FOM University of Applied Sciences for Economics and Management, Germany, with a research focus comparing the sustainability of CNC machining against metal additive manufacturing.

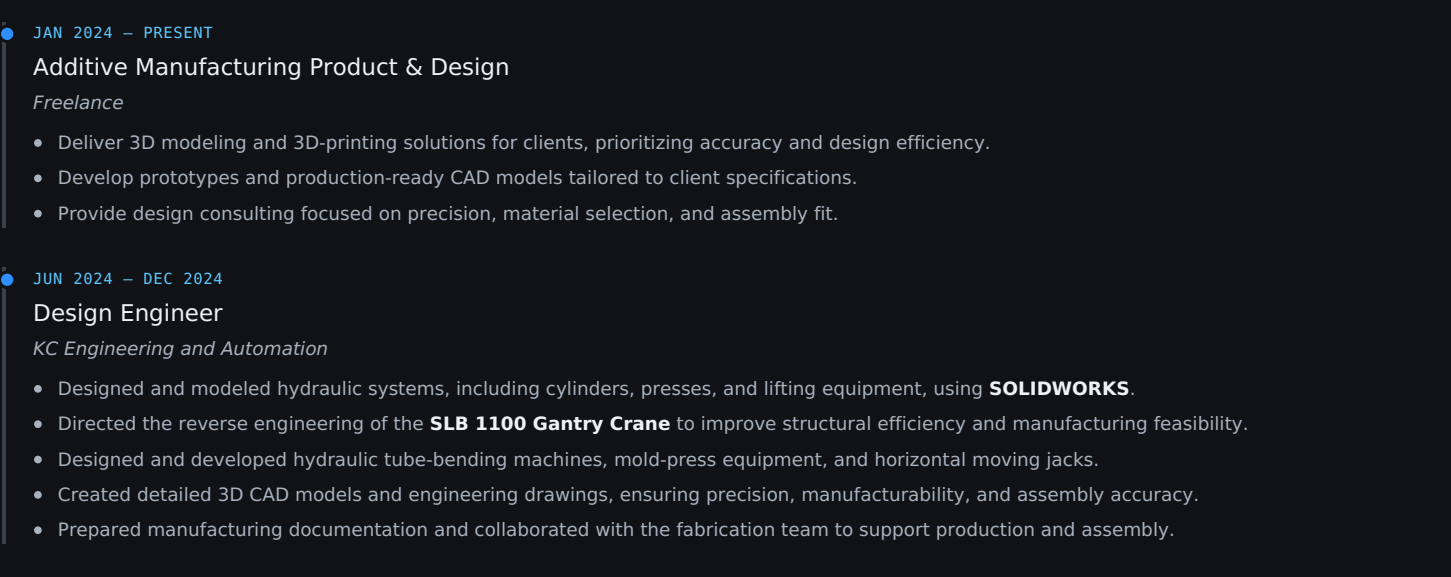
Previously earned a B.Tech in Mechanical Engineering from Symbiosis Institute of Technology, Pune, where hands-on project work included leading the design of Formula Student race car components and hydraulic systems for industrial applications.

Comfortable moving between CAD platforms — **SOLIDWORKS**, **CATIA** and **AutoCAD** — and validating designs in **ANSYS**, with practical Additive Manufacturing experience spanning Cura and RD Works workflows. Tolerances held as tight as 5 microns on production fixture work.

EN – English (C1) HI – Hindi (B1) MR – Marathi (B1) DE – German (A1, pursuing A2)

02 / TOOLSET

Software & capability



03 / EXPERIENCE

Project & work history

- JAN 2024 – PRESENT**

Additive Manufacturing Product & Design

Freelance

 - Deliver 3D modeling and 3D-printing solutions for clients, prioritizing accuracy and design efficiency.
 - Develop prototypes and production-ready CAD models tailored to client specifications.
 - Provide design consulting focused on precision, material selection, and assembly fit.
- JUN 2024 – DEC 2024**

Design Engineer

KC Engineering and Automation

 - Designed and modeled hydraulic systems, including cylinders, presses, and lifting equipment, using **SOLIDWORKS**.
 - Directed the reverse engineering of the **SLB 1100 Gantry Crane** to improve structural efficiency and manufacturing feasibility.
 - Designed and developed hydraulic tube-bending machines, mold-press equipment, and horizontal moving jacks.
 - Created detailed 3D CAD models and engineering drawings, ensuring precision, manufacturability, and assembly accuracy.
 - Prepared manufacturing documentation and collaborated with the fabrication team to support production and assembly.
- JAN 2024 – MAY 2024**

Design Intern

ASB International Pvt. Ltd.

 - Created 3D models of mold components and designed precision fixtures for industrial machinery.
 - Maintained tolerances of **5 microns** to ensure assembly stability during machining.
 - Supported the design team in drafting engineering documentation and production drawings.
- 2022 – 2023**

Design Head

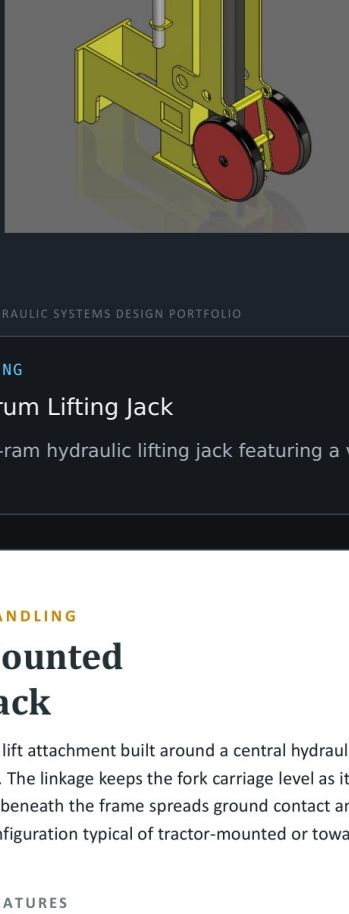
Wrench Welders Racing – Formula Student Team, SIT

 - Led the design and fabrication of Formula Student race car components using **SOLIDWORKS** and **CATIA**.
 - Designed and manufactured carbon-fibre and fibreglass body panels for improved vehicle performance.
 - Developed a high-efficiency EV battery pack, contributing to a **15th-place finish among 72 competing teams**.
 - Built an engine sound signature segregation system using audio analysis and machine learning for vehicle diagnostics.
 - Represented the team at **SUPRA 2022** and **Formula Student EV 2021**.

04 / PORTFOLIO

CAD & hydraulic systems

01



MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

MATERIAL HANDLING

Container / Drum Lifting Jack

A wheeled, single-ram hydraulic lifting jack featuring a vertical mast with dual U-shaped fork saddles for safely lifting and transporting drums or containers.

KEY DESIGN FEATURES

- Dual U-fork saddles for handling loads at two different heights
- Single hydraulic ram, guided along a bolted vertical mast
- Twin large-diameter load wheels for rolling transport once lifted
- Bolted/pinned frame joints for maintenance and part replacement

800 lb
TYPICAL CAPACITY
CLASS

~130 mm
TYPICAL LIFT TRAVEL

2-wheel
ROLLING BASE

Reference specifications based on commercial 55-gal drum/container jack classes (e.g. Global Industrial, Vexel).

01

MATERIAL HANDLING

Fork-Mounted Floor Jack

A parallelogram lift attachment built around a central hydraulic cylinder and two boom arms. The linkage keeps the fork carriage level as it rises, while a sliding-rail base beneath the frame spreads ground contact and adds tipping stability — a configuration typical of tractor-mounted or towable pallet-fork lifters.

KEY DESIGN FEATURES

- Central hydraulic cylinder driving a parallelogram boom linkage
- Fork carriage remains level throughout the lift (no manual tilt correction)
- Sliding-rail/skid base for stability and low ground clearance
- Clevis-pinned joints at every pivot for easy field service

~1–2.5 t
TYPICAL FORK
CAPACITY

Parallel
LINKAGE GEOMETRY

Skid base
STABILITY FEATURE

Reference ranges compiled from common hydraulic pallet-fork and tractor-attachment lift classes.

MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

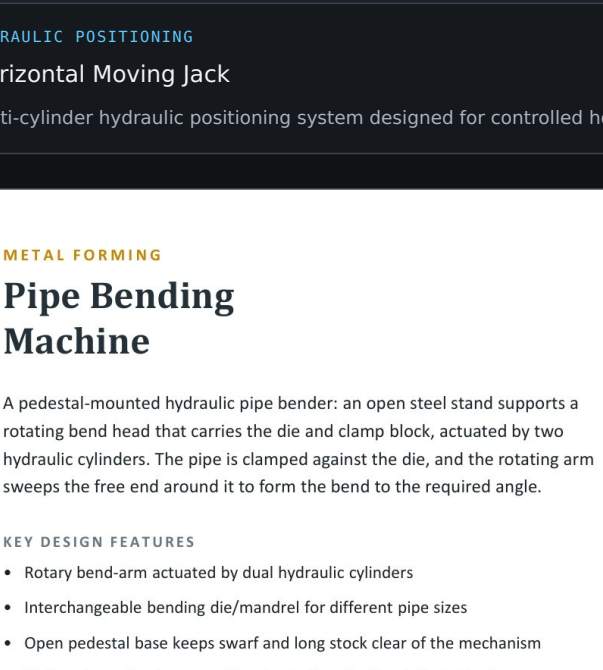
MATERIAL HANDLING

Fork-Mounted Floor Jack

Compact hydraulic fork-mounted lifting system with a parallelogram linkage that maintains level lifting throughout the operating range.

04

03



MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

HYDRAULIC POSITIONING

Horizontal Moving Jack

Multi-cylinder hydraulic positioning system designed for controlled horizontal movement of heavy machinery and industrial equipment.

KEY DESIGN FEATURES

- Three hydraulic cylinders sharing one bolted housing
- Twin lifting-ear brackets for crane handling of the whole unit
- Compact footprint suited to confined installation locations
- Synchronized ram strokes for controlled incremental travel

3 rams
CYLINDER
CONFIGURATION

Multi-ton
SKID/JACK CLASS

Sync.
ACTUATION
STRATEGY

General reference context for multi-cylinder hydraulic sliding/jacking units used in heavy-equipment relocation.

03

METAL FORMING

Pipe Bending Machine

A pedestal-mounted hydraulic pipe bender; an open steel stand supports a rotating bend head that carries the die and clamp block, actuated by two hydraulic cylinders. The pipe is clamped against the die, and the rotating arm sweeps the free end around it to form the bend to the required angle.

KEY DESIGN FEATURES

- Rotary bend-arm actuated by dual hydraulic cylinders
- Interchangeable bending die/mandrel for different pipe sizes
- Open pedestal base keeps swarf and long stock clear of the mechanism
- Sliding clamp block secures the pipe before the bend stroke begins

6–100 t
TYPICAL HYDRAULIC
CLASS

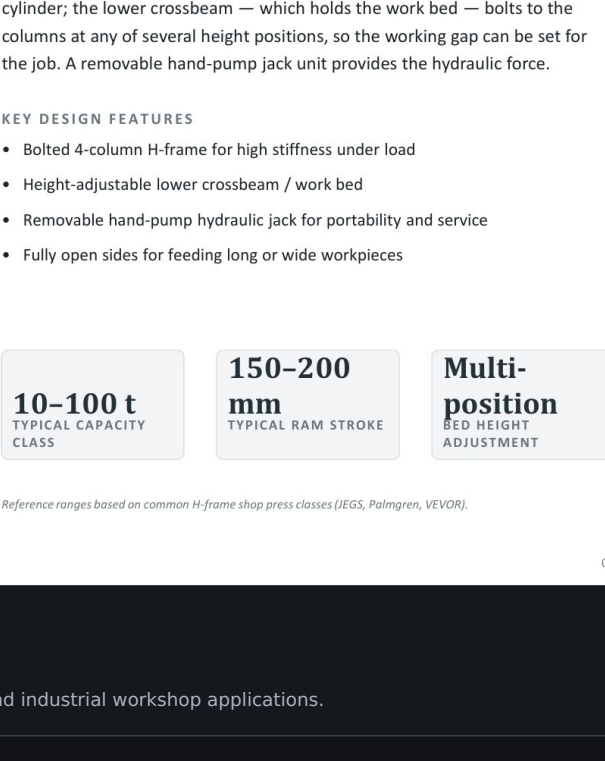
½–6 in
PIPE OD RANGE (BY
CLASS)

Up to 180°
ADJUSTABLE BEND
ANGLE

Reference ranges compiled from commercial hydraulic pipe/hube bender specifications (Hacker Freight, Hines Bending, Rush Machine Tools).

MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

04



04

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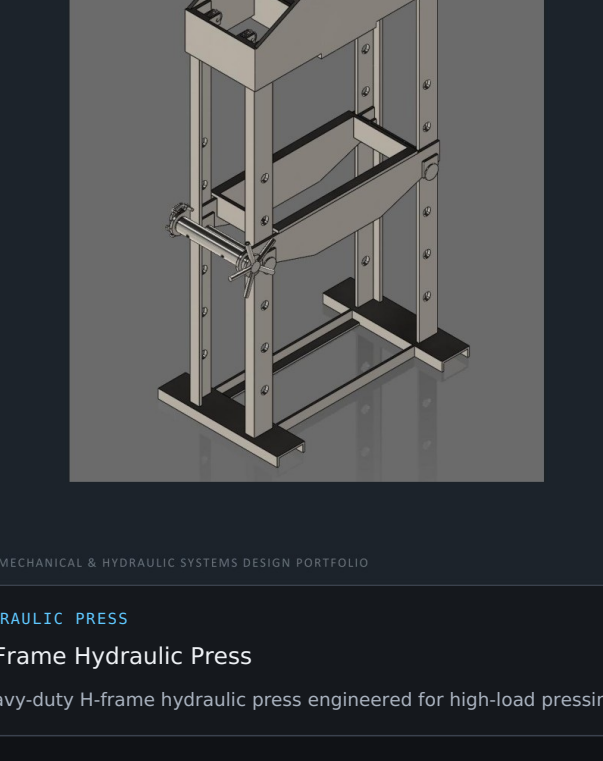
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CLASS

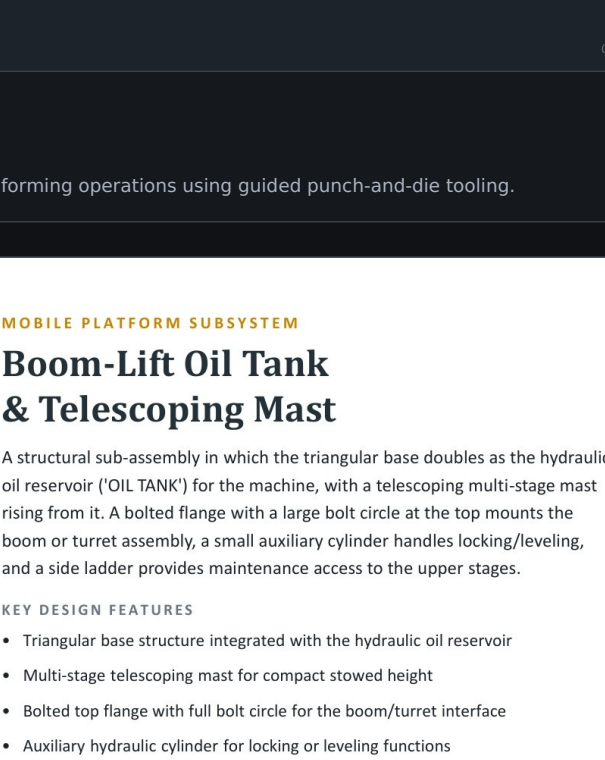
½–6 in
PIPE OD RANGE (BY
CLASS)

Up to 180°
ADJUSTABLE BEND
ANGLE

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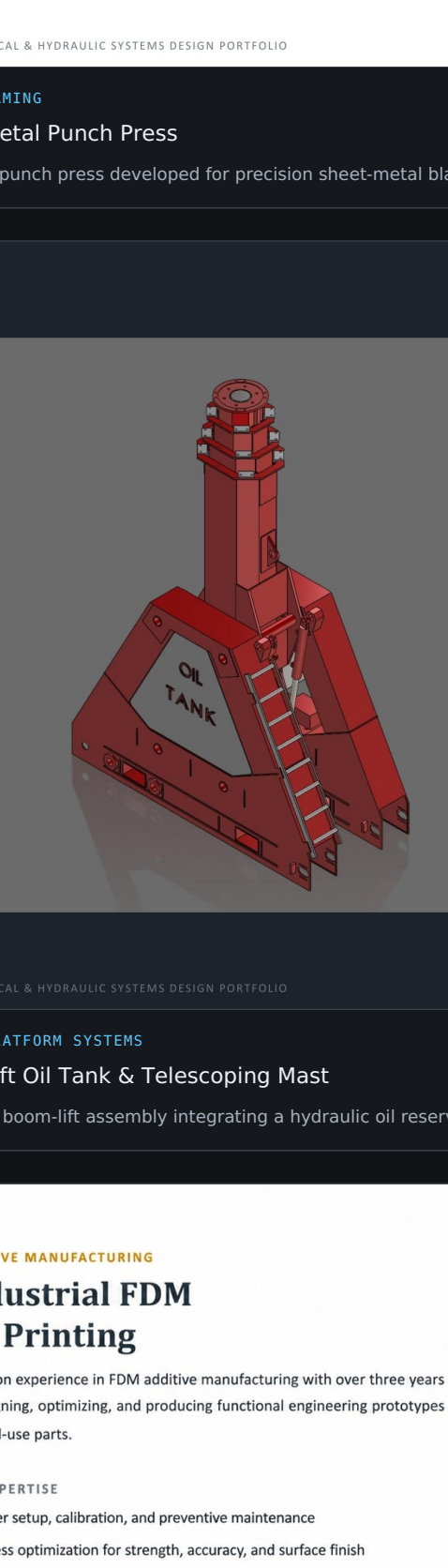
MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

04



04

05



MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

HYDRAULIC PRESS

H-Frame Hydraulic Press

Heavy-duty H-frame hydraulic press engineered for high-load pressing, forming and industrial workshop applications.

KEY DESIGN FEATURES

- Bolted 4-column H-frame for high stiffness under load
- Height-adjustable lower crossbeam / work bed
- Removable hand-pump hydraulic jack for portability and service
- Fully open sides for feeding long or wide workpieces

10–100 t
TYPICAL CAPACITY
CLASS

150–200 mm
TYPICAL RAM STROKE

Multi-position
BED HEIGHT
ADJUSTMENT

Reference ranges based on common H-frame shop press classes (ACES, Palganges, VEXON).

01

METAL FORMING

Sheet Metal Punch Press

The same H-frame press format, fitted here with a matched punch-and-die tooling set for blanking or piercing sheet metal. Guide pins on the die shoe keep the punch aligned to the die opening on every stroke, and the two-piece bolster (upper platen + lower die shoe) spreads the punching load evenly into the frame columns.

KEY DESIGN FEATURES

- Matched punch/die tooling set with guide-pin alignment
- Two-piece bolster plate (platen + die shoe) for even load transfer
- Guided ram rails prevent punch side-load and tooling wear
- Frame shared with the general-purpose press for tooling flexibility

20–150 t
TYPICAL CLASS FOR
BLANKING

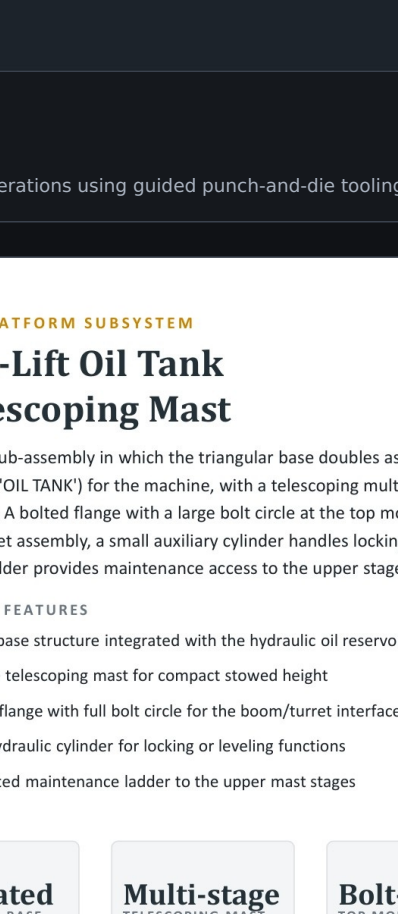
Up to ~6–10 mm
MILD STEEL
THICKNESS

Guided
PUNCH/DIE
ALIGNMENT

Reference ranges for hydraulic punch/blanking press classes on H-frame architectures.

MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

06



06

METAL FORMING

Sheet Metal Punch Press

Hydraulic punch press developed for precision sheet-metal blanking, piercing and forming operations using guided punch-and-die tooling.

KEY DESIGN FEATURES

- Matched punch/die tooling set with guide-pin alignment
- Two-piece bolster plate (platen + die shoe) for even load transfer
- Guided ram rails prevent punch side-load and tooling wear
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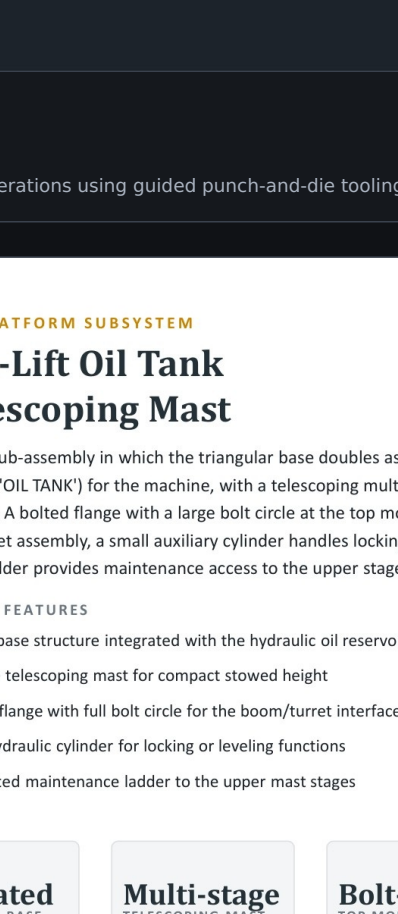
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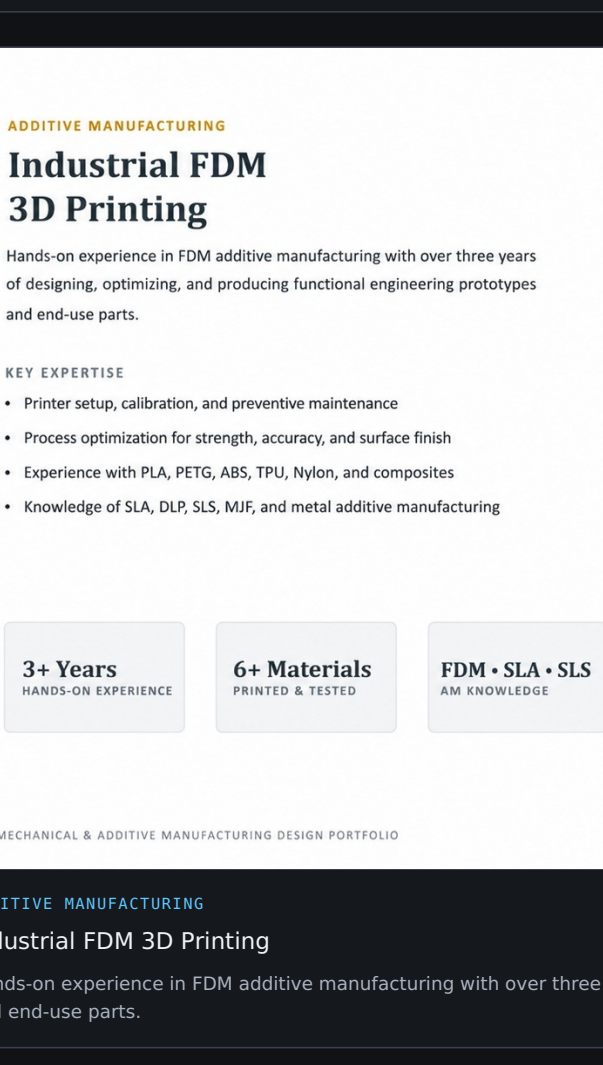
MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

06



06

07



MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

MOBILE PLATFORM SUBSYSTEMS

Boom-Lift Oil Tank & Telescoping Mast

Structural boom-lift assembly integrating a hydraulic oil reservoir with a multi-stage telescoping mast for elevated work platform applications.

KEY DESIGN FEATURES

- Triangular base structure integrated with the hydraulic oil reservoir
- Multi-stage telescoping mast for compact stowed height
- Bolted top flange with full bolt circle for the boom/turret interface
- Auxiliary hydraulic cylinder for locking or leveling functions
- Side-mounted maintenance ladder to the upper mast stages

Integrated
RESERVOIR + BASE

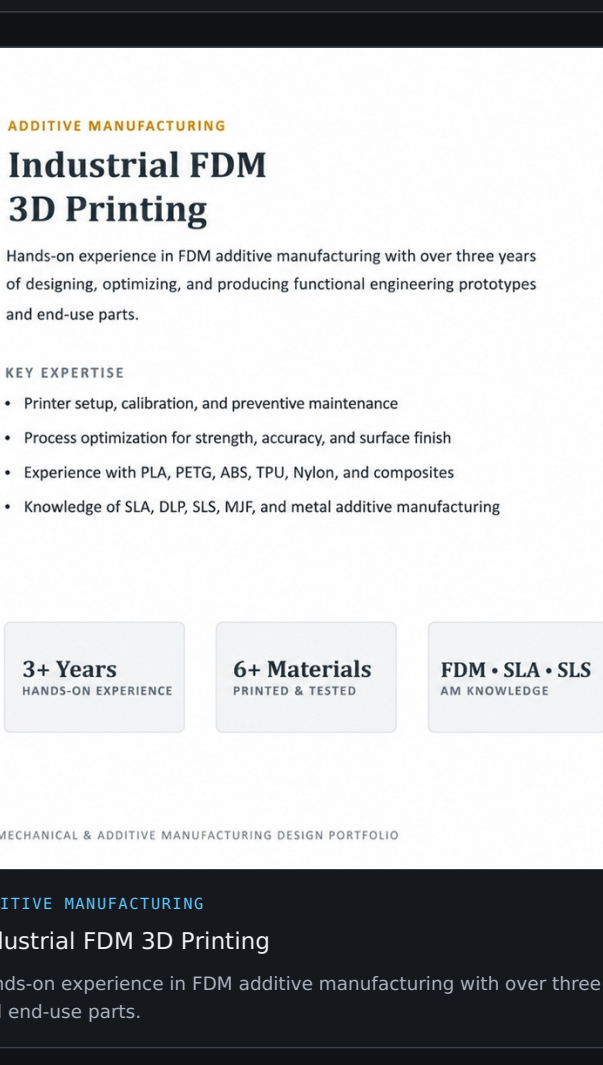
Multi-stage
TELESCOPING MAST

Bolt-circle
TOP MOUNTING
FLANGE

Functional description based on the modeled geometry. Typical of mobile elevating work platform (MEWP) main/reservoir subassemblies.

01

07



MECHANICAL & HYDRAULIC SYSTEMS DESIGN PORTFOLIO

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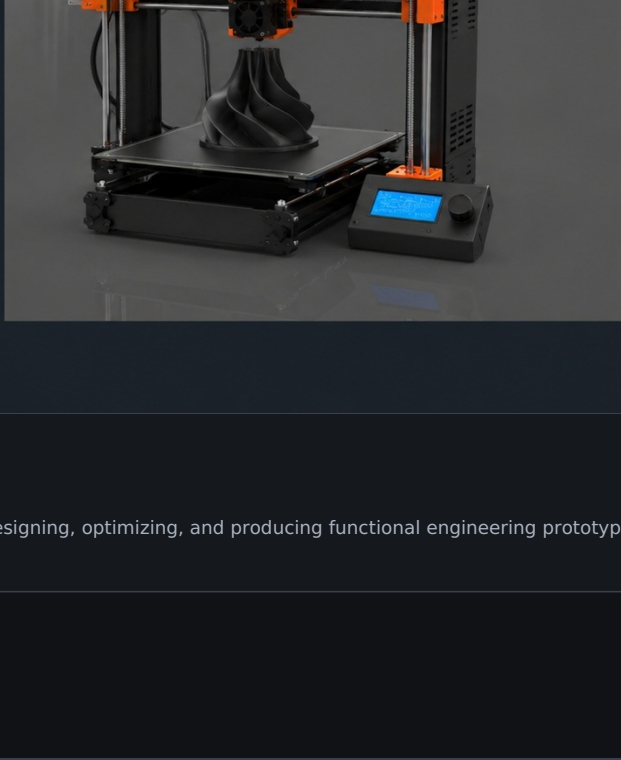
Multi-stage
TELESCOPING MAST

Bolt-circle
TOP MOUNTING
FLANGE

Functional description based on the modeled geometry. Typical of mobile elevating work platform (MEWP) main/reservoir subassemblies.

01

08



08

ADDITIVE MANUFACTURING

Industrial FDM 3D Printing

Hands-on experience in FDM additive manufacturing with over three years of designing, optimizing, and producing functional engineering prototypes and end-use parts.

KEY EXPERTISE

- Printer setup, calibration, and preventive maintenance
- Process optimization for strength, accuracy, and surface finish
- Experience with PLA, PETG, ABS, SLS, TPU, Nylon, and composites
- Knowledge of SLA, DLP, SLS, MJF, and metal additive manufacturing

3+ Years
HANDS-ON EXPERIENCE

6+ Materials
PRINTED & TESTED

FDM • SLA • SLS
AM KNOWLEDGE

MECHANICAL & ADDITIVE MANUFACTURING DESIGN PORTFOLIO

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AM KNOWLEDGE

MECHANICAL & ADDITIVE MANUFACTURING DESIGN PORTFOLIO

05 / EDUCATION

Academic record

- APR. 2025 – APR. 2027**

Master of Science in Digital Technologies

FOM Hochschule für Oekonomie & Management, Germany
- JUN. 2020 – JUN. 2024**

Bachelor of Technology in Mechanical Engineering

Symbiosis Institute of Technology, Pune — CGPA 7.0 / 10
- FEB. 2018 – FEB. 2019**

Higher Secondary Certificate, Science

Fr. Agnel's Multipurpose School and Jr. College
- MAR. 2016 – MAR. 2017**

Secondary School Certificate Standard

Fr. Agnel's Multipurpose School and Jr. College

06 / CONTACT

Let's build something precise

Open to product design roles, freelance CAD work and collaboration on hydraulic systems or additive manufacturing projects.

Email nirajnaidu0411@gmail.com

LinkedIn [linkedin.com/in/niraj11](https://www.linkedin.com/in/niraj11)

Phone +49 15757055980

PORTFOLIO ARCHIVE

Research Papers

COMPLETED · JUL 2026

Sustainable Mold Design with Additive Manufacturing

This research explores sustainable approaches to injection mold manufacturing using additive manufacturing technologies. It demonstrates how optimized design techniques, including lightweight structures and topology optimization, can reduce material consumption, lower production costs, and improve manufacturing efficiency while supporting environmentally responsible production.

ADDITIVE MANUFACTURING · TOPOLOGY OPTIMIZATION · SUSTAINABILITY

[VIEW RESEARCH PAPER →](#)

COMPLETED · FEB 2026

Advanced Conformal Cooling for Injection Molding

This study examines the integration of hybrid manufacturing and conformal cooling technologies to enhance injection molding performance. It highlights how advanced cooling channel designs improve heat transfer, reduce cycle times, increase product quality, and deliver greater manufacturing efficiency.

INJECTION MOLDING · CONFORMAL COOLING · HYBRID MANUFACTURING

[VIEW RESEARCH PAPER →](#)

COMPLETED · FEB 2026

Additive vs. CNC Manufacturing: A Performance Comparison

This comparative research evaluates additive manufacturing and traditional CNC machining for injection mold production. It analyzes material efficiency, production time, cost, design flexibility, and overall performance to identify effective manufacturing approaches for different industrial applications.

CNC MACHINING · 3D PRINTING · COMPARISON STUDY

[VIEW RESEARCH PAPER →](#)

COMPLETED · JUL 2026

AI-Driven Quality Control in Additive Manufacturing

This research investigates the use of artificial intelligence, machine learning, and digital twin technologies for predictive quality control in additive manufacturing. It explores real-time defect detection, intelligent process monitoring, and autonomous quality assurance techniques for future smart digital factories.

ARTIFICIAL INTELLIGENCE · DIGITAL TWIN · SMART FACTORY

[VIEW RESEARCH PAPER →](#)

PORTFOLIO ARCHIVE

Certifications

ISSUED · JUL 2026

MINT-AE Study Orientation — 3D Printing

Certificate of Participation from the MINT-AE study orientation program, "From Idea to Reality through 3D Printing," held in the 3D Printing Laboratory of the Faculty of Chemistry, University of Duisburg-Essen (Co Creation Lab Product Innovations). Covered digital model development and prototype production in the 3D printing lab. Issued 10 Jul 2026 by FOM Hochschule für Oekonomie & Management in cooperation with zdi-Zentrum MINT-Netzwerk Essen and the University of Duisburg-Essen.

3D PRINTING · DIGITAL MODELING · PROTOTYPING

[VIEW CERTIFICATE →](#)

ISSUED · JUL 2023

IELTS Certificate

English language proficiency certification (International English Language Testing System). Academic module. Overall Band Score 7.0 (CEFR C1) — Listening 7.5, Reading 7.0, Writing 6.0, Speaking 7.0.

ENGLISH PROFICIENCY · OVERALL BAND 7.0

[VIEW CERTIFICATE →](#)

ISSUED · SEP 2023

Essential in CATIA

Certificate of Completion from CADD Centre Training Services, Mumbai — a 48-hour program covering core CATIA workflows including surface modeling, generative drafting, sketch creation, part modeling, assembly modeling, and sheet metal modeling. Completed 29 Jul 2023 – 31 Aug 2023.

SURFACE MODELING · PART MODELING · ASSEMBLY MODELING

[VIEW CERTIFICATE →](#)

ISSUED · SEP 2023

Diploma in Product Design — Proficient

Certificate of Completion from CADD Centre Training Services, Mumbai, covering AutoCAD drafting and documentation alongside SolidWorks part, assembly, surface, and sheet metal design. Program duration: 11 Dec 2019 – 28 Feb 2020.

AUTOCAD · SOLIDWORKS · TECHNICAL DRAFTING

[VIEW CERTIFICATE →](#)